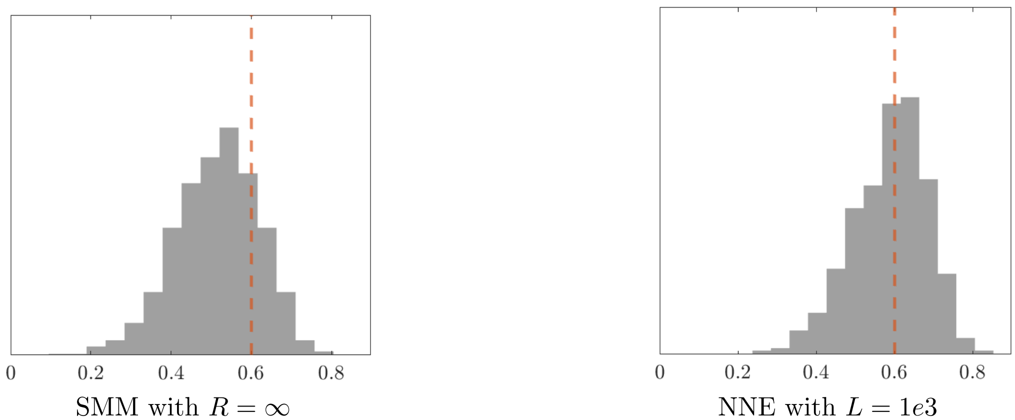

Notes: Histograms of the estimates of β across 1000 Monte Carlo datasets. The vertical dashed red lines display the true value of β. The moment specification follows row 6 in Table 2.

Figure 3: Distributions of Estimates for AR(1) with Redundant Moments

$\beta^k/(1-\beta^2)$, so each of these added moments by itself is informative about β. However, moments with more lags (i.e., larger k) may become almost redundant after we include moments with fewer lags (i.e., smaller k). In both of these rows, GMM has larger bias and RMSE than the benchmark case (row 1). In comparison, NNE's performance remains close to the benchmark case.

Row 5 and 6 add higher-order moments to the benchmark case. While these moments may appear informative at glance, a bit of algebra will show that $\mathbb{E}(y_i^2 y_{i-k}) = 0$ and $\mathbb{E}(y_i y_{i-k}^2) = 0$ regardless of β. Therefore, these moments are irrelevant and redundant. In both rows, GMM now has much larger bias and RMSE than the benchmark case. In comparison, NNE's performance remains close to the benchmark case. For a visual comparison, Figure 3 plots the histograms of the estimates using the nine moments in row 6. The bias of GMM is clear.

To summarize, Table 2 shows that NNE is less sensitive to redundant moments, consistent with our discussion in Section 2. Note, in particular, that the simplicity of AR(1) allows us to derive closed-form expressions of moments and see whether a moment may be redundant. However, doing so more generally in structural estimation is not feasible. In applications of GMM/SMM, researchers often need to carefully select moments rather than just "throwing in" moments that might or might not be useful for recovering parameters. In this regard, the robustness of NNE lessens the burden on researchers to select moments.

4 Application to Consumer Sequential Search

This section studies an application to the sequential search model (Weitzman, 1979, Ursu 2018). The model has attracted growing interests as data on consumer online purchase journey become increasingly available (i.e., what alternative options consumers viewed before purchases). The model is typically estimated by SMLE and the estimation is known to be difficult. The main